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BALLARI INSTITUTE OF TECHNOLOGY & MANAGEMENT

AUTONOMOUS INSTITUTE UNDER VISVESVARAYA TECHNOLOGICAL
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Department Of Computer Science & Engineering Data Science

Seminar Report on

“Advanced Data Visualization Techniques using Power BI”

For the course:

Data Visualization – 22CD633

Submitted by

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Introduction to Security Visualization using Power BI

Security Visualization refers to the process of representing security-related data in the form of charts, dashboards, graphs, and reports to improve monitoring and analysis. Power BI is widely used for security analytics because it helps organizations identify threats, detect unusual activities, and monitor system performance in real time.

By converting raw security data into interactive dashboards, Power BI enables security teams to make faster and better decisions. It provides an easy way to analyze logs, alerts, user activities, and network performance visually.

Power BI Workflow for Security Visualization

1. Data Collection

Security data is collected from multiple sources such as firewalls, servers, antivirus software, cloud platforms, and network devices.

2. Data Transformation

The collected data is cleaned, filtered, and transformed using Power Query Editor to remove unwanted information and prepare it for analysis.

3. Data Modeling

Relationships are created between different datasets to improve analysis and reporting.

4. Data Visualization

Interactive dashboards and visuals are created to display security insights such as attacks, threats, login failures, and suspicious activities.

5. Monitoring and Sharing

The dashboards are published to Power BI Service for continuous monitoring and collaboration among security teams.

Features of Security Visualization in Power BI

Power BI provides several important features for security analysis:

- Real-time security monitoring
- Interactive dashboards
- Threat analysis and detection
- Easy identification of suspicious activities
- Integration with multiple security data sources
- Detailed reports and alerts
- User-friendly interface

These features help organizations improve cybersecurity management and decision-making.

Security Dashboard

A Security Dashboard is used to monitor and analyze security-related information visually. It displays important metrics and alerts in one place for quick understanding.

The dashboard may include:

- Number of cyber attacks
- Failed login attempts
- Malware detection reports
- Network traffic analysis
- User activity monitoring
- Security risk levels

Security dashboards help administrators quickly identify threats and take necessary actions.

Charts Used in Security Visualization

1. Bar Chart

Used to compare different security incidents such as malware attacks, phishing attempts, and login failures.

2. Line Chart

Used to show trends in security events over time.

3. Pie Chart

Used to display the percentage distribution of different attack types.

4. Map Visualization

Used to identify attack locations and geographical threat distribution.

5. KPI Cards

Used to display important values such as total attacks detected, blocked threats, or active alerts.

Advantages of Security Visualization

- Faster detection of threats
- Better understanding of security risks
- Improved decision-making
- Real-time monitoring of systems
- Efficient incident management
- Enhanced cybersecurity awareness

Visualization helps security teams analyze large amounts of data easily and respond quickly to potential attacks.

Applications of Security Visualization

Security visualization is widely used in:

- Network security monitoring
- Cloud security analysis
- Banking and financial security
- Government cybersecurity systems
- Healthcare data protection
- IT infrastructure monitoring

Organizations use security dashboards to improve system protection and reduce cyber risks.

Design Tips for Security Dashboards

- Use simple and clear visuals

- Highlight critical alerts properly
- Avoid unnecessary information
- Use proper titles and labels
- Ensure easy readability of dashboards
- Focus on important security metrics

A well-designed dashboard improves the effectiveness of security analysis.

Conclusion

Security Visualization using Power BI helps organizations monitor and analyze cybersecurity data effectively. Interactive dashboards and advanced visuals enable faster identification of threats, unusual activities, and security risks.

Power BI provides powerful tools for real-time monitoring, reporting, and analysis, making it highly useful for cybersecurity management. By using proper visualization techniques, organizations can improve security operations and make better strategic decisions against cyber threats.